Design and Implementation
Line Impedance Stabilization Network (LISN)

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Chapter 1 Introduction

1.1 Document Objectives

The Line Impedance Stabilization Network (LISN) is an indispensable device in the evaluation of conducted electromagnetic emissions from electronic systems. It is used to provide a measurement environment with a defined impedance while isolating the noise generated by the equipment under test from the noise present on the power supply network. As a result, the measurement results exhibit high repeatability and can be compared across different test sessions or laboratories.

The objective of this document is to present the complete design and implementation process of a laboratory LISN for research purposes. The content includes the theoretical background, design requirements, component selection methodology, hardware implementation, calibration, and the operating procedures for conducted emission measurements.

In addition to describing the fabricated device, this document also aims to preserve the design decisions and calculation basis to facilitate future maintenance, improvements, and the development of subsequent LISN versions.

1.2 Introduction to Electromagnetic Compatibility

The rapid development of modern electronic systems has led to a significant increase in sources of electromagnetic interference. Switching power supplies, inverters, high-speed microcontrollers, motor control circuits, wireless communication devices, and various power semiconductor components are all capable of generating electromagnetic noise during operation.

If left uncontrolled, this interference can propagate to other equipment through conductive paths or radiate into free space, resulting in degraded performance, data corruption, system instability, or even damage to sensitive electronic devices.

To ensure that electronic devices can operate simultaneously within the same environment without interfering with one another, the concept of **Electromagnetic Compatibility (EMC)** was introduced. EMC is defined as the ability of a device to operate correctly in its intended electromagnetic environment without generating electromagnetic disturbances that exceed acceptable limits for other equipment.

EMC consists of two primary aspects:

- **Electromagnetic Interference (EMI):** the generation and emission of electromagnetic disturbances by a device.
- **Electromagnetic Susceptibility (EMS):** the ability of a device to continue operating correctly when exposed to external electromagnetic disturbances.

A device is considered EMC compliant only when both conditions are satisfied: its emissions remain within the specified limits, and it possesses adequate immunity to electromagnetic disturbances from the surrounding environment.

During electronic product development, EMC evaluation is typically performed from the research and design stages to identify electromagnetic emission issues early, thereby reducing the cost of modifications before formal EMC certification.

1.3 Conducted Electromagnetic Emissions

Electromagnetic interference generated by electronic equipment can propagate through different mechanisms. Based on the propagation medium, electromagnetic emissions are generally classified into two main categories: Conducted Emission and Radiated Emission.

Conducted emission refers to electromagnetic noise that propagates through conductive paths connecting equipment to the power supply or interconnecting different devices. Since power cables unintentionally act as transmission paths, the noise can travel over long distances and affect multiple devices connected to the same electrical system.

In most residential and industrial EMC standards, conducted emission measurements are performed over the frequency range from $150\,\mathrm{kHz}$ to $30\,\mathrm{MHz}$. This frequency range is where conductive paths have a significant influence on electromagnetic noise propagation and also represents the primary operating range of the LISN presented in this document.

Conducted emission evaluation is typically carried out during product development to identify the primary noise sources, optimize EMI filters, and verify the effectiveness of noise mitigation techniques before conducting formal EMC certification tests.

1.4 Role of the LISN in Conducted Emission Measurements

In practice, the impedance of the power supply network is not constant but depends on many factors, including cable length, power system configuration, connected loads, and the characteristics of the power source. Furthermore, the power network itself contains electromagnetic noise generated by other electrical equipment connected to the same system.

If the equipment under test is connected directly to the power supply for conducted emission measurements, the measurement results will depend significantly on the electrical network conditions at the time of testing. This reduces measurement repeatability and makes it difficult to compare results across different test sessions or laboratories.

The Line Impedance Stabilization Network was developed to address this problem. A LISN provides a standardized impedance between the power supply and the equipment under test

while isolating the measurement equipment from noise present on the power network. In addition, it provides a standardized measurement port for coupling high-frequency noise to a spectrum analyzer or an EMI receiver.

By standardizing the impedance of the power supply network, conducted emission measurements achieve high repeatability and conform to EMC evaluation procedures used in both research and industrial applications.

The operating principles and detailed structure of the LISN are presented in Chapter 2.

1.5 Scope of the Document

This document focuses on the design and implementation of a laboratory LISN for the evaluation of conducted emissions from low-power electronic systems, switching power supplies, and embedded systems.

The document covers the theoretical background, design methodology, hardware implementation, calibration, and operating instructions for the device. Topics related to EMC certification procedures performed by accredited testing laboratories are introduced only briefly and are not the primary focus of this document.

Chapter 2 Theoretical Background

2.1 Conducted Electromagnetic Emissions

In modern electronic systems, particularly switching mode power supplies (SMPS), inverters, motor controllers, and high-speed digital circuits, the presence of electromagnetic interference is unavoidable. Power semiconductor devices such as MOSFETs, IGBTs, and GaN transistors operate by switching at very high speeds, producing voltage (dv/dt) and current (di/dt) transitions with high slew rates. This switching process is the primary source of electromagnetic interference in power electronic systems.

According to Fourier analysis, a switching signal contains not only the fundamental switching frequency but also numerous harmonic components at higher frequencies. Although a power supply may operate at only several tens or hundreds of kilohertz, its noise spectrum can extend to tens or even hundreds of megahertz. Therefore, evaluating the spectral characteristics of the generated noise is an important step in system design and optimization.

Electromagnetic interference can propagate through different mechanisms. Based on the propagation medium, it is generally classified into two main categories:

- **Conducted Emission**: Interference that propagates through conductors such as power cables, signal cables, or ground connections.
- **Radiated Emission**: Interference that is emitted as electromagnetic waves and propagates through free space.

The scope of this document focuses on Conducted Emission, as it is the primary type of interference that the Line Impedance Stabilization Network (LISN) is designed to evaluate.

For Conducted Emission, power cables no longer serve solely as a means of transmitting electrical power but also act as propagation paths for high-frequency noise components. These noise components can travel from the Device Under Test (DUT) back toward the power source or propagate to other equipment connected to the same electrical system.

According to EMC standards such as CISPR 16 and CISPR 25 [2, 3], Conducted Emission measurements are typically performed over the frequency range from $150\,\mathrm{kHz}$ to $30\,\mathrm{MHz}$. This is the frequency range in which conducted interference is dominant and also the primary operating range of the LISN presented in this document.

Accurate determination of the conducted emission level enables designers to evaluate the effectiveness of EMI filters, identify the primary emission sources, and optimize the hardware design before conducting formal EMC certification tests.

2.2 Noise Propagation Modes

Differential Mode and Common Mode Noise

Characteristic	Differential Mode (DM) Noise	Common Mode (CM) Noise
Definition	Noise appears in **opposite phase** between two signal conductors (or between a signal conductor and its return/local ground).	Noise appears **in phase** with approximately equal amplitude on both conductors (with respect to a common ground reference).
Current Path	The noise current flows through one conductor and **returns** through the other conductor, forming a closed current loop.	The noise current flows along both signal conductors and **returns through the ground (GND) path** or parasitic ground paths.
Effect on the Circuit	Directly affects the differential voltage ($V_{in+} - V_{in-}$), resulting in direct distortion of the useful signal.	Affects both inputs simultaneously and, in theory, is rejected by circuits with a high common-mode rejection ratio (CMRR).

2.3 Challenges in Conducted Emission Measurements

The objective of Conducted Emission measurement is to accurately determine the level of high-frequency noise generated by the DUT and propagated back toward the power source. However, under practical conditions, performing this measurement is not straightforward because the power supply system itself exhibits electrical characteristics that vary over time.

The impedance of the power supply network depends on many factors, including cable length, power network configuration, connected loads, connector contact quality, and the characteristics of the power source. Consequently, the same device may produce different measurement results when tested at different locations or at different times.

Furthermore, the power network always contains various sources of interference, such as switching power supplies, electric motors, inverters, and other electronic equipment connected to the same electrical system. These interference components may overlap with the signal being measured, making it difficult to accurately determine the noise generated by the DUT.

Another issue is that a spectrum analyzer or EMI receiver is typically designed with a standard input impedance of $50\,\Omega$. If it is connected directly to the power line, the measurement equipment is not only exposed to the risk of damage from the mains voltage but also alters the electrical characteristics of the system, causing the measurement results to no longer represent the actual operating condition of the DUT.

For these reasons, a Conducted Emission measurement requires an intermediate network that both provides isolation between the power source and the measurement equipment and establishes a measurement environment with a defined and stable impedance throughout the test. The device that performs this function is the Line Impedance Stabilization Network (LISN), which will be discussed in detail in the following sections.

2.4 Introduction to the Line Impedance Stabilization Network

As discussed in Section 2.3, the impedance of a practical power supply system is not a fixed quantity. It varies depending on the power network configuration, cable length, power source characteristics, and the loads operating on the same system. Therefore, if the Device Under Test (DUT) is connected directly to the power supply for Conducted Emission measurements, the measurement results will depend significantly on the condition of the power source at the time of testing.

To eliminate this dependency, an impedance stabilization network is placed between the power source and the DUT. This network is called the **Line Impedance Stabilization Network (LISN)**.

A LISN is a measurement device used in Conducted Emission testing to provide an electrical environment with defined and repeatable characteristics. Instead of allowing the DUT to operate directly from a power source with unknown impedance, the LISN provides a standardized impedance over the entire measurement frequency range, ensuring that the measurement results are not affected by variations in the power supply system.

In addition to impedance stabilization, the LISN also separates the low-frequency power signal from the high-frequency noise components generated by the DUT. This allows the DUT to operate normally from either an AC or DC power source while coupling the RF noise components to a spectrum analyzer or EMI receiver for measurement.

The LISN can be regarded as an interface between the three components of the measurement system:

1. The power source supplying energy to the DUT.
2. The Device Under Test (DUT).
3. The measurement instrument, typically a Spectrum Analyzer or EMI Receiver.

During operation, the LISN must satisfy several technical requirements simultaneously. First, it must allow power-frequency current (50 Hz or 60 Hz) to pass to the DUT with minimal attenuation so that the operating conditions of the DUT remain unchanged. Second, the high-frequency noise components generated by the DUT must be directed toward the RF measurement port rather than propagating back to the power source. Finally, the measurement port must provide an impedance compatible with the measurement equipment, which is typically standardized at $50\,\Omega$ in RF measurement systems.

By satisfying these requirements simultaneously, the LISN isolates the influence of the power network on Conducted Emission measurements while enabling highly repeatable testing. For this reason, the LISN is considered a standard component in most EMC measurement systems, ranging from laboratory research setups to formal compliance testing based on international standards.

Depending on the applicable test standard and the intended application, the structure of a LISN may differ in terms of inductance value, operating frequency range, current rating, or

connection configuration. Nevertheless, all LISN variants are based on the same fundamental principle: standardizing the impedance of the power line and separating the high-frequency noise components from the power signal.

2.5 Operating Principle of the Line Impedance Stabilization Network

Having understood the role of the LISN in a Conducted Emission measurement system, this section analyzes its operating principle. Although the detailed structure of a LISN may vary depending on the applicable standard or the power range of the DUT, all LISN configurations operate based on the same principle: **separating the power transmission path from the high-frequency noise transmission path**.

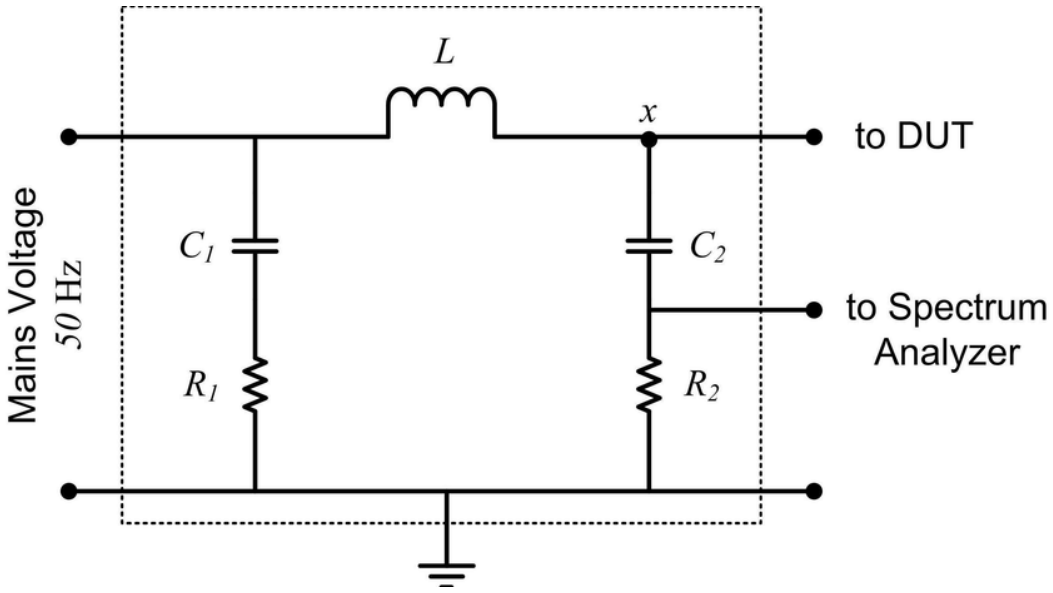


Figure 2.1: Basic schematic of the Line Impedance Stabilization Network.

Figure 2.1 shows the basic schematic of the LISN. The device is placed between the power source and the DUT while providing an RF port for connection to a Spectrum Analyzer or EMI Receiver. Although the circuit may include various protection and coupling components, the operating principle of the entire system is primarily based on the frequency-dependent characteristics of inductors and capacitors.

First, consider the mains power supply operating at $50\,\mathrm{Hz}$. At this frequency, the impedance of the inductor is very small compared to the load impedance. Consequently, power is transferred almost directly from the source to the DUT, producing only a negligible voltage drop across the LISN. This ensures that inserting the LISN into the measurement setup does not significantly alter the operating conditions of the Device Under Test.

In contrast, the noise generated by the DUT typically lies within the frequency range from several hundred kilohertz to several tens of megahertz. As the frequency increases, the impedance of the inductor increases according to

$$X_L = 2\pi f L \tag{2.1}$$

Meanwhile, the capacitor has an impedance given by

$$X_C = \frac{1}{2\pi f C} \tag{2.2}$$

This means that as the frequency increases, the inductor gradually acts as an obstacle to the noise current propagating back toward the power source, while the capacitor provides a low-impedance path for the high-frequency components.

As a result of the combination of these two components, the power current and the noise current are naturally separated into different paths. The power-frequency current continues to flow to the DUT to supply energy, whereas the high-frequency noise current is directed to the RF measurement port instead of propagating back into the power network.

To ensure compatibility with RF measurement equipment, the measurement port of the LISN is standardized to an impedance of $50\,\Omega$. This is the standard impedance widely used in Spectrum Analyzers, EMI Receivers, and Vector Network Analyzers. Maintaining this standard impedance not only reduces signal reflections but also ensures that measurement results can be compared across different instruments and laboratories.

It can be seen that a LISN does not operate as a conventional EMI filter. An EMI filter is designed to prevent interference from propagating into the surrounding environment, whereas the purpose of a LISN is to provide a measurement environment with a defined impedance and a standardized path for capturing the noise generated by the DUT. For this reason, the measured noise level obtained using a LISN may, in many cases, be higher than that obtained when the DUT is connected directly to the power supply. However, only the former accurately represents the emission characteristics of the DUT under standardized measurement conditions.

2.6 Analysis of the Main Components of a LISN

Although various LISN configurations exist depending on the applicable test standard and the power rating of the Device Under Test, most Line Impedance Stabilization Networks consist of several basic functional blocks. Each component performs a specific function in standardizing the power-line impedance, isolating noise from the power network, and providing a path for the noise signal to reach the measurement instrument.

This section analyzes the role of each component based on the schematic presented in Section 2.5.

2.6.1 Inductor

The inductor is the most important component in the structure of a LISN. Its primary function is to create a difference in impedance between the power frequency and the frequency range of the noise signal.

For the mains power signal at $50\,\mathrm{Hz}$, the inductive reactance is very small. Therefore, almost all of the power from the source is delivered to the DUT without altering its operating conditions.

In contrast, as the frequency increases into the range of several hundred kilohertz or several tens of megahertz, the inductive reactance increases linearly with frequency. This limits the propagation of the noise components toward the power source and forces the high-frequency energy to follow the intended path within the LISN.

$$X_L = 2\pi f L \tag{2.3}$$

In LISN design, the inductance value is not selected arbitrarily but is specified by the corresponding EMC measurement standards. This value determines the impedance characteristics of the entire system over the measurement frequency range and directly affects the accuracy of Conducted Emission measurements.

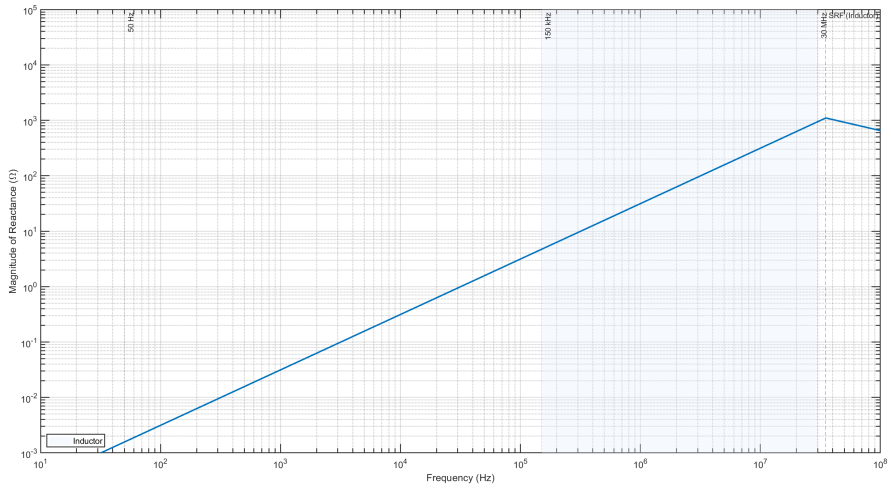


Figure 2.2: Variation of inductive reactance with frequency for the LISN inductor.

2.6.2 Coupling Capacitor

The coupling capacitor is used to couple the high-frequency noise components from the power line to the RF measurement port while preventing the power supply voltage from reaching the measurement instrument.

The characteristics of a capacitor are opposite to those of an inductor. The impedance of a capacitor decreases as the frequency increases according to

$$X_C = \frac{1}{2\pi f C} \tag{2.4}$$

At the power frequency, the capacitive reactance is very high, so virtually no power current flows through the measurement branch. However, within the Conducted Emission measurement frequency range, the impedance of the capacitor decreases significantly, providing a low-impedance path for the noise signal to the Spectrum Analyzer.

The selection of the capacitance value must balance the ability to transmit high-frequency signals with the requirement of isolating the power supply voltage. In LISN designs, the coupling capacitor is typically a safety capacitor or a capacitor with a high voltage rating to ensure safe operation.

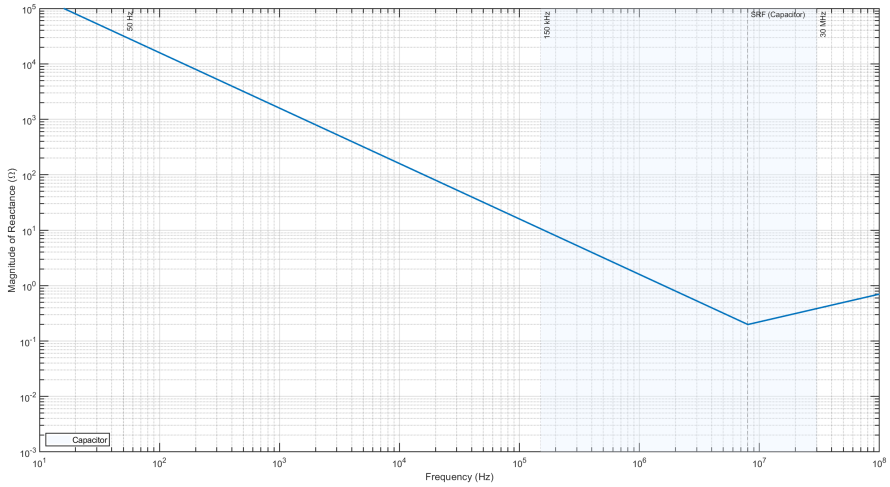


Figure 2.3: Variation of capacitive reactance with frequency for the coupling capacitor in the LISN.

2.6.3 Standard $50\,\Omega$ Impedance Network

RF measurement instruments such as Spectrum Analyzers, EMI Receivers, and Vector Network Analyzers are designed with a standard input impedance of $50\,\Omega$. Therefore, the measurement port of the LISN must also maintain the same impedance to ensure compatibility among the instruments in the measurement system.

If the impedance at the measurement port is not standardized, the noise signal may be reflected at the connection point, resulting in measurement errors in the spectrum amplitude and reducing the repeatability of the measurement.

In addition to impedance matching, the resistor network contributes to establishing the standardized impedance seen by the DUT over the entire measurement frequency range. This is one of the most important factors that allows Conducted Emission measurement results to be compared across different laboratories.

2.7 LISN Configurations

Although all LISNs share the common purpose of standardizing the source impedance and providing a measurement port for conducted emissions, various configurations exist depending on the test object and the applicable EMC standard. These configurations differ in topology, inductance value, and the number of supported power lines.

2.7.1 Classification by Topology

Based on circuit topology, LISNs are generally classified into three main types:

- **V-LISN (Voltage LISN):** This is the most commonly used configuration for Conducted Emission measurements. A V-LISN measures the noise voltage between each power conductor and ground. The measured value includes both Common Mode and Differential Mode noise components.
- **Delta LISN:** A Delta LISN provides two separate outputs for Common Mode and Differential Mode noise. In practice, the same functionality can also be achieved by combining two V-LISNs with a signal separator (LISN Mate).
- **T-ISN (Telecommunication ISN):** Unlike the previous two configurations, a T-ISN is designed for data transmission lines or telecommunication cables rather than power lines. It is primarily used to measure the Common Mode component on twisted-pair cables.

2.7.2 Classification by Inductance Value

Another classification method is based on the inductance value of the impedance stabilization network. The two most commonly used configurations are the $5\,\mu\mathrm{H}$ LISN and the $50\,\mu\mathrm{H}$ LISN.

Table 2.1: Comparison of Two Common LISN Configurations

Characteristic	$5\,\mu\mathrm{H}$ **LISN**	$50\,\mu\mathrm{H}$ **LISN**
Application	Automotive electronic equipment	Commercial and industrial electronic equipment
Simulates	Short power cable (approximately 5 m)	Power wiring in building electrical systems
Typical frequency range	150 kHz – 110 MHz	9 kHz – 30 MHz
Typical standard	CISPR 25	CISPR 16

According to the Tekbox documentation, the $5\,\mu\mathrm{H}$ LISN is primarily used for EMC testing of automotive electronic equipment, whereas the $50\,\mu\mathrm{H}$ LISN is widely used for equipment operating in commercial and industrial environments. In addition to having different inductance values, the two configurations also differ in their applicable frequency ranges and corresponding standard requirements.

2.7.3 Classification by the Number of Power Lines

LISNs can also be classified according to the number of power lines integrated into a single device.

- **Single-path LISN**: Used for a single DC power line or a single conductor under test.
- **Two-path LISN**: Commonly used for DC power systems consisting of positive and negative terminals, or single-phase AC power systems consisting of Line and Neutral conductors.
- **Four-path LISN**: Used for three-phase power systems, including three phase conductors and one neutral conductor.

Within the scope of this document, the designed LISN is a single-path V-LISN with a nominal inductance of $5\,\mu\mathrm{H}$. This configuration is suitable for Conducted Emission research on DC-powered electronic systems and complies with the requirements of CISPR 25.

Chapter 3 Hardware Design

After presenting the fundamental principles of Conducted Emission measurements and the structure of the Line Impedance Stabilization Network (LISN) in Chapter 2, this chapter focuses on the hardware design process of the LISN developed within the scope of this document.

This chapter analyzes the design requirements, component selection criteria, and the factors affecting the performance of the LISN over the measurement frequency range. Based on these considerations, each functional block of the system is presented, including the impedance stabilization network, the RF coupling circuit, the protection circuit, and the PCB design requirements.

3.1 Design Requirements

Designing a Line Impedance Stabilization Network (LISN) involves more than simply selecting component values according to a schematic. The device must also satisfy electrical, mechanical, and measurement requirements appropriate for its intended application. These requirements form the basis for selecting the circuit configuration, components, and hardware implementation presented in the following sections.

Within the scope of this document, the LISN is developed for laboratory research and the evaluation of Conducted Emission from DC-powered electronic systems. The design emphasizes ease of fabrication, reasonable cost, and straightforward calibration while maintaining the essential characteristics required for pre-compliance measurements.

The main design requirements are defined as follows:

- Use a V-LISN configuration with a nominal inductance of $5\,\mu\mathrm{H}$, suitable for Conducted Emission measurements of DC-powered electronic systems.
- Operate over the frequency range from $150\,\mathrm{kHz}$ to $110\,\mathrm{MHz}$, meeting the requirements for CISPR 25 pre-compliance measurements.
- Maintain a standard $50\,\Omega$ impedance at the RF port to ensure direct compatibility with a Spectrum Analyzer or EMI Receiver.
- Allow power current to be supplied to the DUT without causing a significant voltage drop along the power line or affecting the normal operating conditions of the Device Under Test.

- Minimize the influence of external power-line noise on the measurement results while coupling the noise generated by the DUT to the measurement port with minimal attenuation.
- Ensure electrical safety through protective components such as fuses, varistors, and bleed resistors to reduce risks to both the user and the measurement equipment.
- Prioritize the use of commercially available components that are readily obtainable, facilitating fabrication and maintenance under laboratory conditions.
- Design the PCB to minimize parasitic inductance and parasitic capacitance while maintaining adequate electrical clearance for the system operating voltage.

Table 3.1: LISN Design Requirements

Parameter	**Value**	**Purpose**
Nominal inductance	5 μH	Standardize the source impedance according to the selected configuration
Operating frequency range	150 kHz – 110 MHz	Conducted Emission measurement
RF port impedance	50 Ω	Compatibility with RF measurement instruments
RF connector type	BNC	Connection to a Spectrum Analyzer
Power supply type	DC	Suitable for the target application
Protection level	Limiter + Filter + Attenuator	Ensure safe operation

3.2 Design of the Impedance Stabilization Network

The impedance stabilization network is the most important functional block of a Line Impedance Stabilization Network (LISN). It determines the impedance seen by the Device Under Test (DUT) during measurement while isolating high-frequency noise from the power source to facilitate signal acquisition at the RF port.

As discussed in Chapter 2, the impedance of the LISN must remain stable over the entire measurement frequency range to ensure the repeatability and consistency of Conducted Emission measurements. Therefore, selecting the inductance value and the structure of the impedance stabilization network is one of the most important design decisions.

Within the scope of this document, the LISN is implemented using a $5\,\mu\mathrm{H}$ configuration. This configuration is widely used for Conducted Emission measurements of DC-powered electronic systems and is consistent with the objectives of this work.

The impedance stabilization network consists of a series inductor placed in the power line together with auxiliary components that ensure the impedance characteristics of the entire system. Among these components, the inductor is the primary element responsible for differentiating the low-frequency power signal from the high-frequency noise components.

At the power frequency, the inductive reactance is very small, so the supply voltage delivered to the DUT is almost unaffected. As the frequency increases into the Conducted Emission measurement range, the inductive reactance increases rapidly with frequency, limiting the propagation of noise toward the power source. Consequently, most of the energy contained in the noise signal is directed to the RF coupling network for measurement.

In addition to the nominal inductance, the performance of the impedance stabilization network is also affected by practical component characteristics such as DC resistance (DCR), self-resonant frequency (SRF), saturation current, and parasitic parameters. These factors must all be considered during component selection to ensure that the impedance characteristics of the LISN are maintained throughout the operating frequency range.

The following sections describe the selection of the inductance value, the criteria for choosing the inductor, and the considerations involved in the hardware implementation.

3.2.1 Selection of the Inductance Value

For Conducted Emission measurements of AC-powered electronic equipment, the $50\,\mu\mathrm{H}$ LISN configuration is commonly used to simulate the impedance of residential or industrial power distribution systems. In contrast, for DC-powered electronic systems, particularly in the automotive field, CISPR 25 specifies the use of a $5\,\mu\mathrm{H}$ LISN configuration. The lower inductance value reflects the characteristics of the short, low-impedance power cables typically found in vehicle electrical systems.

Within the scope of this document, the LISN is developed for laboratory research and Conducted Emission evaluation of DC-powered electronic systems. Therefore, the $5\,\mu\mathrm{H}$ configuration is selected as the design basis to ensure compatibility with the CISPR 25 measurement methodology and the intended application of the device.

Table 3.2: Reasons for Selecting the $5\,\mu\mathrm{H}$ LISN Configuration

Criterion	**Reason for Selection**
Measurement target	DC-powered electronic equipment
Reference standard	CISPR 25
Nominal inductance	$5\,\mu\mathrm{H}$
Operating frequency range	$150\,\mathrm{kHz}$ – $110\,\mathrm{MHz}$
Objective	Laboratory Conducted Emission measurement

3.2.2 Inductor Selection

After determining the nominal inductance value of $5\,\mu H$, the next step is to select the implementation method for the inductor. Two common approaches were considered: using a commercial inductor and fabricating a custom air-core inductor.

Air-core inductors are commonly used in EMC measurement equipment due to their linear characteristics, the absence of magnetic saturation, and negligible core losses. In addition, the designer can adjust the number of turns, coil diameter, and turn spacing to optimize the inductance value as well as the Self-Resonant Frequency (SRF).

However, this approach requires suitable wire, fabrication tools, and measurement equipment to calibrate the inductance after winding. For the prototype developed in this work, fabricating a custom inductor would significantly increase both the implementation time and the equipment requirements.

In contrast, commercially available power inductors are offered in a wide range of $5\,\mu H$ values, with high current ratings and operating frequency characteristics suitable for Conducted Emission measurements. The use of commercial components reduces manufacturing time, ensures repeatability between different versions, and simplifies replacement when necessary.

Table 3.3 presents a comparison between the two approaches.

Table 3.3: Comparison of Two Inductor Implementation Approaches

Criterion	**Commercial Inductor**	**Custom-Wound Inductor**
Implementation time	Short	Longer
Repeatability	High	Depends on the fabrication process
Parameter optimization	Limited	Highly flexible
Equipment requirements	Minimal	Requires fabrication and measurement equipment
Replaceability	Easy	Must be fabricated again
Suitability for MVP	High	Moderate

Commercial Inductor Approach

The commercial inductor approach was selected to reduce design time, simplify the fabrication process, and ensure repeatability between different test iterations. Modern power inductors are manufactured using industrial production processes, provide complete technical specifications, and have verified current-handling capability and frequency characteristics, making them suitable for a laboratory LISN intended for research purposes.

In this design, a Würth Elektronik WE-HCC series power inductor, part number 7443340100, was selected as the reference component. The inductor uses a ferrite core, a wirewound structure, and is manufactured as a Surface Mount Device (SMD). Five inductors are connected in series to obtain a total inductance of $5\,\mu H$.

The main specifications of the selected component are listed in Table 3.4.

Table 3.4: Specifications of the Selected Commercial Inductor

Parameter	Value
Nominal inductance	$1\,\mu\mathrm{H}$
Tolerance	$\pm 20\%$
Continuous current	$17\,\mathrm{A}$
Saturation current	$24\,\mathrm{A}$
DC resistance (DCR)	$3.9\,\mathrm{m}\Omega$ (Max.)
Self-Resonant Frequency (SRF)	$147\,\mathrm{MHz}$
Core material	Ferrite
Package type	SMD

Among these specifications, the Self-Resonant Frequency (SRF) is one of the most important criteria when selecting an inductor for a LISN. With an SRF of approximately $147\,\mathrm{MHz}$, which is higher than the upper limit of the Conducted Emission measurement range ($110\,\mathrm{MHz}$), the inductor maintains its inductive behavior throughout the entire operating frequency range. This minimizes the influence of parasitic capacitance and preserves the characteristics of the impedance stabilization network.

In addition, the DC resistance of only approximately $3.9\,\mathrm{m}\Omega$ minimizes the voltage drop along the power line and reduces power dissipation as the load current increases. The continuous current rating of up to $17\,\mathrm{A}$ also satisfies the requirements of the system during the research and testing stages.

Although ferrite-core inductors may experience magnetic saturation under high current or overcurrent pulse conditions, the saturation current rating of $24\,\mathrm{A}$ is significantly higher than the expected operating current of the system. Therefore, the effect of magnetic saturation under normal operating conditions is considered negligible.

Custom Air-Core Inductor Approach

In addition to using a commercial inductor, another approach considered was the fabrication of a custom air-core inductor. This method is commonly used in RF and EMC equipment when frequency characteristics need to be optimized or when the required inductance value is not commercially available.

Unlike ferrite-core inductors, air-core inductors do not exhibit magnetic saturation and have no magnetic core losses. As a result, the inductance remains stable over the entire operating current range, while the frequency response is primarily determined by the coil geometry and its parasitic parameters.

In the proposed design, the inductor is wound using copper wire with a diameter of approximately $1.6\,\mathrm{mm}$. The wire is wound around a former fabricated using 3D printing technology

to ensure that the coil diameter, turn spacing, and coil length remain consistent throughout the manufacturing process.

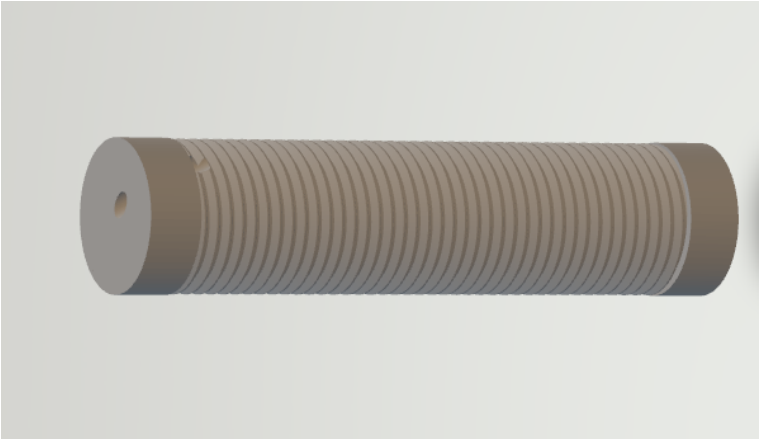


Figure 3.1: Proposed design of the 3D-printed coil former.

The inductance value of the coil is determined using empirical equations for air-core inductors. One of the most widely used equations is Wheeler's formula [4]:

$$L(\mu H) = \frac{r^2 N^2}{9r + 10l} \tag{3.1}$$

where

- L is the inductance of the coil (μH);
- r is the mean radius of the coil (inch);
- l is the coil length (inch);
- N is the number of turns.

This equation provides an initial estimate of the required number of turns before fabrication. After winding, the actual inductance is measured using an LCR meter and adjusted, if necessary, by changing the number of turns or the spacing between adjacent turns.

One notable advantage of this approach is that the designer can optimize the geometric parameters of the coil to increase the Self-Resonant Frequency (SRF) and reduce the parasitic capacitance between turns. This is particularly beneficial for Conducted Emission measurements, where the frequency response of the inductor directly affects the characteristics of the impedance stabilization network.

However, this approach also has several disadvantages. The fabrication process requires appropriate mechanical tools, measurement equipment for post-fabrication inductance adjustment, and the repeatability between different coils depends on the manufacturing accuracy.

Furthermore, the physical size of an air-core inductor is generally much larger than that of a commercial inductor with the same inductance value.

3.3 Design and Simulation

Evaluating the performance of a Line Impedance Stabilization Network (LISN) based solely on component specifications is insufficient to verify compliance with the requirements of CISPR 25. In practice, the response of a LISN is influenced by the combined effects of the inductive network, capacitors, load resistance, parasitic elements, and the interactions among all components within the system.

Therefore, before hardware fabrication, the design was verified through SPICE circuit simulations. The simulations focused on evaluating the most important performance parameters of the LISN, including output impedance, insertion loss, isolation between the power source and the DUT, and the influence of parasitic parameters on the frequency response of the system.

Unlike the previous sections, which analyzed individual components separately, the simulations presented in this chapter were performed on the complete LISN schematic to closely represent actual operating conditions. After confirming that the basic structure satisfies the design requirements, additional circuits such as the attenuator, limiter, and filter are incorporated and evaluated in the following sections.

3.3.1 Simulation Model

The simulations were performed using the complete $5\,\mu\mathrm{H}$ LISN model intended for DC power systems. At this stage, the model includes only the core components of a standard LISN, without additional protection or signal-conditioning circuits such as the attenuator, limiter, or filter. This approach allows the intrinsic characteristics of the impedance stabilization network to be evaluated before integrating the additional functional blocks.

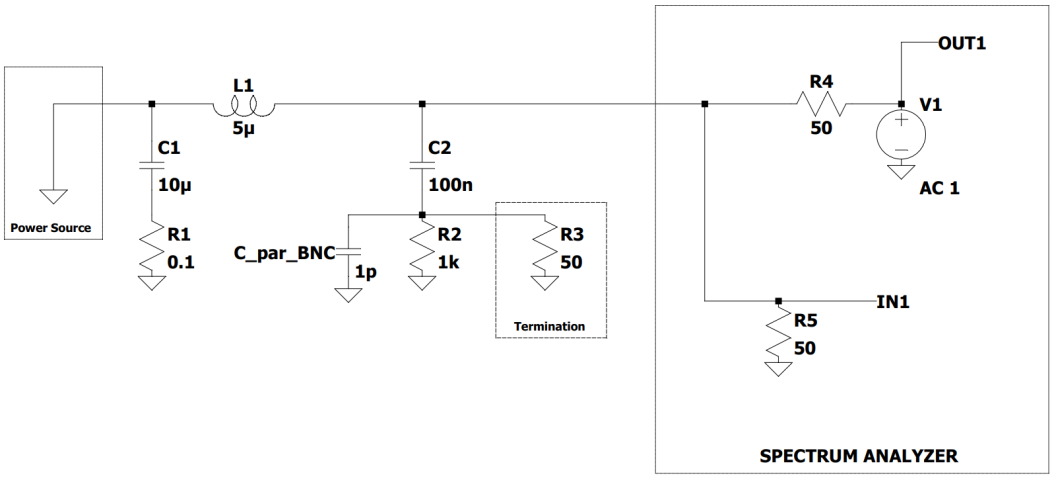


Figure 3.2: Simulation model of the LISN used for performance evaluation.

In this model, the power supply is assumed to be an ideal voltage source with negligible internal impedance. The Device Under Test (DUT) is represented by an equivalent load together with an AC source for frequency-domain analysis. The RF output of the LISN is terminated with a standard $50\,\Omega$ load, corresponding to the input impedance of a Spectrum Analyzer or EMI Receiver.

To comprehensively evaluate the performance of the LISN, the following simulations were carried out:

1. Output impedance simulation;
2. Insertion loss simulation;
3. Input-to-output isolation simulation.

This approach makes it possible to verify each key performance parameter of the LISN before adding the input protection circuits. Consequently, any changes in the frequency response caused by integrating the attenuator, limiter, or filter can be evaluated independently and directly compared with the baseline LISN configuration.

3.3.2 Output Impedance Simulation

The output impedance is the most important parameter of a Line Impedance Stabilization Network (LISN), as it determines the ability to provide a standardized measurement environment for Conducted Emission testing. According to the requirements of CISPR 25, the impedance seen from the DUT port must remain at a specified value throughout the entire measurement frequency range to ensure repeatable and comparable results across different laboratories.

Since practical components exhibit parasitic elements and frequency-dependent characteristics, evaluating the impedance of a LISN solely through theoretical calculations is insufficient. Therefore, before hardware fabrication, the impedance characteristics of the complete circuit were verified through AC frequency-domain simulation.

Simulation Setup

The simulation was performed using LTspice [5] based on the complete LISN schematic shown in Figure 3.3. The model includes the $5\,\mu\mathrm{H}$ inductor, bypass capacitors, damping resistors, and the standard load in order to accurately represent the behavior of the actual circuit.

The Spectrum Analyzer input is modeled as a standard $50\,\Omega$ impedance. An AC voltage source with an amplitude of $1\,\mathrm{V}$ is used to excite the circuit while performing a frequency sweep from $10\,\mathrm{kHz}$ to $200\,\mathrm{MHz}$. In addition to the primary components, the parasitic capacitance of the BNC connector is included to evaluate the influence of non-ideal elements at high frequencies.

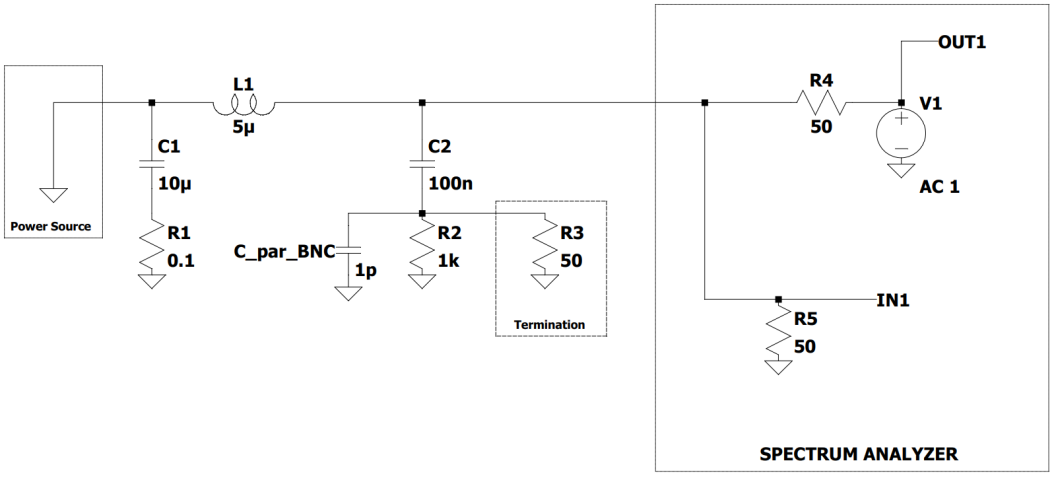


Figure 3.3: Simulation setup used to determine the output impedance of the LISN.

In this model, the output impedance is not calculated directly from the voltage-to-current ratio but is derived from the reflection coefficient of the $50\,\Omega$ system. With both the source and the load matched to $50\,\Omega$, the output impedance is calculated as

$$Z_{out} = 50 \cdot \frac{\dfrac{V_{IN}}{V_{OUT}}}{1 - 2\dfrac{V_{IN}}{V_{OUT}}} \tag{3.2}$$

where V_{OUT} is the excitation voltage and V_{IN} is the voltage measured at the LISN measurement port. This expression is implemented directly in LTspice to generate the output impedance versus frequency characteristic.

Results and Discussion

The simulation results are presented as the impedance magnitude versus frequency in Figure 3.4. Within the operating frequency range, the impedance of the LISN remains close to the intended design value while varying continuously according to the response of the LC network.

At higher frequencies, the parasitic capacitances of the inductor, connectors, and PCB layout begin to influence the system response, producing the Self-Resonant Frequency (SRF). Beyond this frequency, the inductor gradually loses its inductive behavior, and the entire network begins to exhibit capacitive characteristics. Therefore, selecting an inductor with a sufficiently high SRF is essential to ensure that the resonance occurs outside the intended operating frequency range of the LISN.

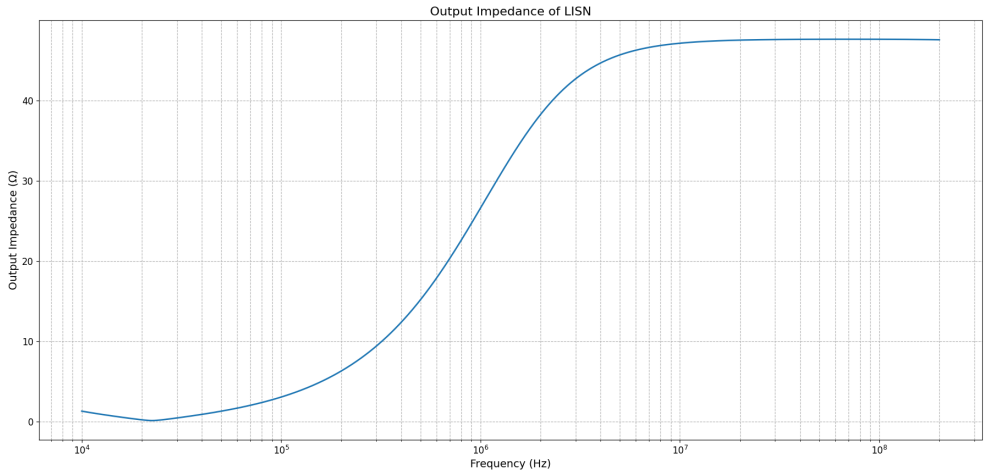


Figure 3.4: Simulated output impedance characteristic of the LISN obtained from AC analysis.

This simulation serves as the initial verification step to confirm the fundamental characteristics of the impedance stabilization network before introducing additional functional blocks such as the attenuator, limiter, and RF filter. These additional circuits are analyzed in the following sections to evaluate their effects on the overall frequency response of the system.

Experimental Verification

After hardware fabrication, the impedance characteristics of the LISN should be experimentally verified to validate the simulation results.

In practice, this measurement is typically performed using a Vector Network Analyzer (VNA) or an Impedance Analyzer after proper $50\,\Omega$ system calibration. The measurement instrument is connected directly to the RF port of the LISN, and a frequency sweep is performed over the entire operating range. From the measured scattering parameter S_{11} or the measured voltage and current values, the output impedance of the LISN can be determined and compared with the simulation results.

Within the scope of this chapter, these procedures are introduced only to complete the overall design methodology. The experimental measurement procedure, including the connection setup, instrument calibration, and data processing methods, will be described in detail in the system implementation chapter.

3.3.3 Insertion Loss Simulation

In addition to maintaining a standardized impedance, the LISN must ensure that the conducted noise generated by the DUT is transmitted to the measurement port with minimal attenuation and stable frequency characteristics. If the insertion loss is excessive or varies significantly with frequency, the Conducted Emission measurement will no longer accurately represent the actual noise level produced by the DUT.

The insertion loss simulation was performed to evaluate the signal transmission capability of the LISN from the DUT port to the RF port. In this simulation model, an AC source is connected to the DUT side, while the RF output is terminated with a standard $50\,\Omega$ load corresponding to the input impedance of a Spectrum Analyzer.

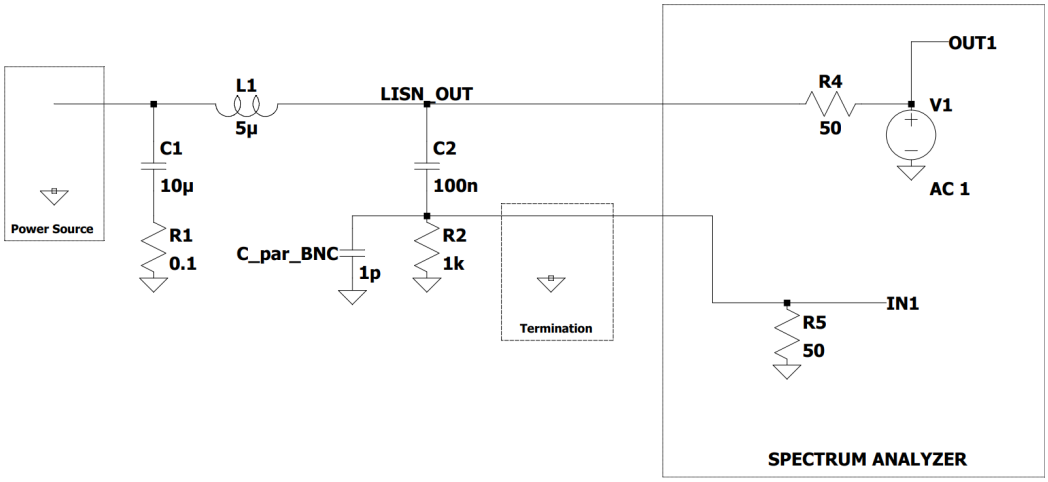


Figure 3.5: Simulation setup for the LISN insertion loss analysis.

The transmission response is determined from the voltage ratio between the RF port and the LISN output:

$$H(f) = 20\log_{10}\left|\frac{V_{RF}}{V_{LISN-OUT}}\right| \tag{3.3}$$

where $V_{LISN-OUT}$ is the voltage at the DUT side and V_{RF} is the voltage measured at the RF port.

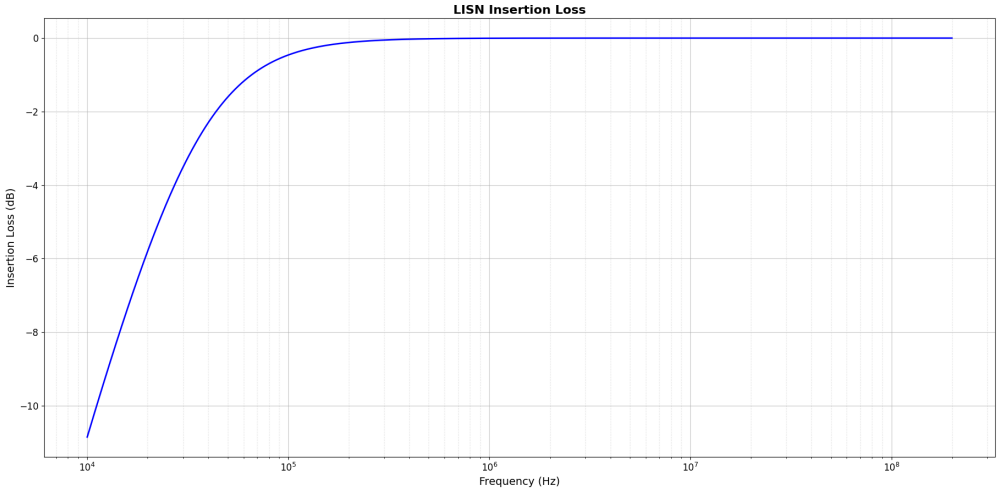


Figure 3.6: Simulated insertion loss response of the LISN.

Within the CISPR 25 measurement frequency range, the simulation results show that the signal is transmitted from the DUT to the RF port with low and relatively constant attenuation.

This indicates that the AC coupling network of the LISN performs as intended, allowing the high-frequency noise components to reach the measurement instrument while preventing the DC supply voltage from appearing at the input of the Spectrum Analyzer.

The transmission response also provides insight into the influence of the coupling components, particularly the RF coupling capacitor and the $50\,\Omega$ termination resistor. Deviations in the response at higher frequencies are mainly caused by the parasitic capacitance of the components, the parasitic inductance of the PCB traces, or the self-resonant frequency limitations of the inductor.

3.3.4 Input-to-Output Isolation Simulation

Besides output impedance and transmission characteristics, another important performance parameter of a LISN is the isolation between the power input and the RF measurement port (Input-to-Output Isolation). This parameter indicates how much noise originating from the power supply can propagate directly to the input of the Spectrum Analyzer. If the isolation is insufficient, supply noise will appear at the measurement port and distort the Conducted Emission results of the DUT.

The Input-to-Output Isolation simulation was performed to evaluate the attenuation provided by the LISN network for signals propagating from the power input to the RF port over the entire operating frequency range.

The simulation setup is shown in Figure 3.7. An AC source with an amplitude of $1\,\mathrm{V}$ is connected to the power input of the LISN, while the RF port is terminated with a standard $50\,\Omega$ load representing the input impedance of the Spectrum Analyzer. The filter network components, including the inductor, coupling capacitor, and parasitic elements, are identical to those used in the impedance simulation model in order to accurately evaluate the influence of practical components.

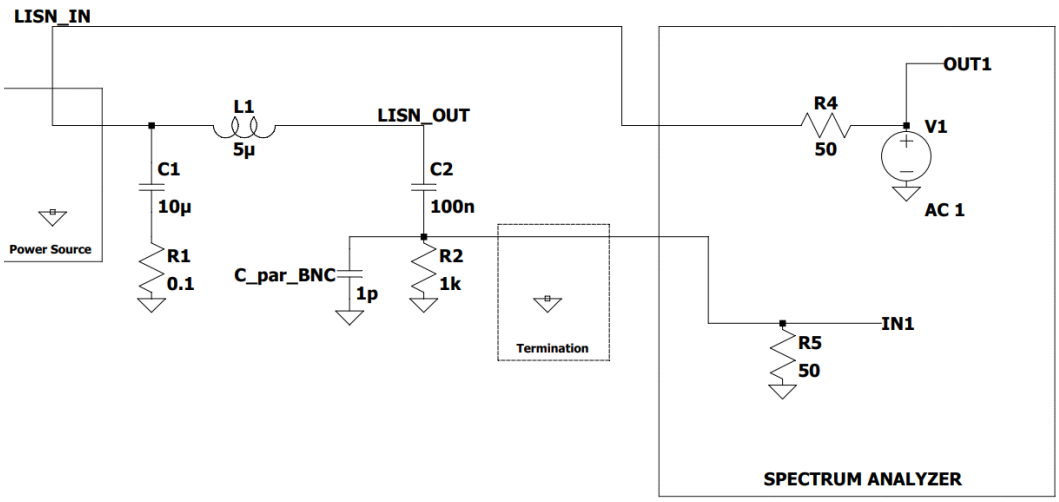


Figure 3.7: Simulation setup for the LISN Input-to-Output Isolation analysis.

The isolation is calculated from the voltage transfer function between the power input and the RF port:

$$Isolation(f) = 20\log_{10}\left|\frac{V_{RF}}{V_{IN}}\right| \tag{3.4}$$

where V_{IN} is the voltage at the LISN power input and V_{RF} is the voltage measured at the RF port.

The simulation results are presented in Figure 3.8. Throughout the CISPR 25 Conducted Emission frequency range, the isolation remains high, with the transfer coefficient remaining at a low level and decreasing further as the frequency increases. This indicates that most of the noise originating from the power source is attenuated by the LC network before reaching the RF port, thereby minimizing the influence of supply noise on EMI measurements.

At higher frequencies, a resonance peak appears due to the inter-winding capacitance of the inductor. As the frequency approaches the Self-Resonant Frequency (SRF), the inductive behavior of the inductor gradually deteriorates, reducing the isolation performance of the filter network.

In this model, the resonance occurs at approximately $150\,\mathrm{MHz}$, which is well above the upper limit of the Conducted Emission measurement range ($30\,\mathrm{MHz}$). Therefore, this resonance does not affect the operating performance of the LISN within the frequency range of interest. Moreover, maintaining the SRF outside the measurement band indicates that the parasitic capacitance of the inductor has been adequately controlled during the design process.

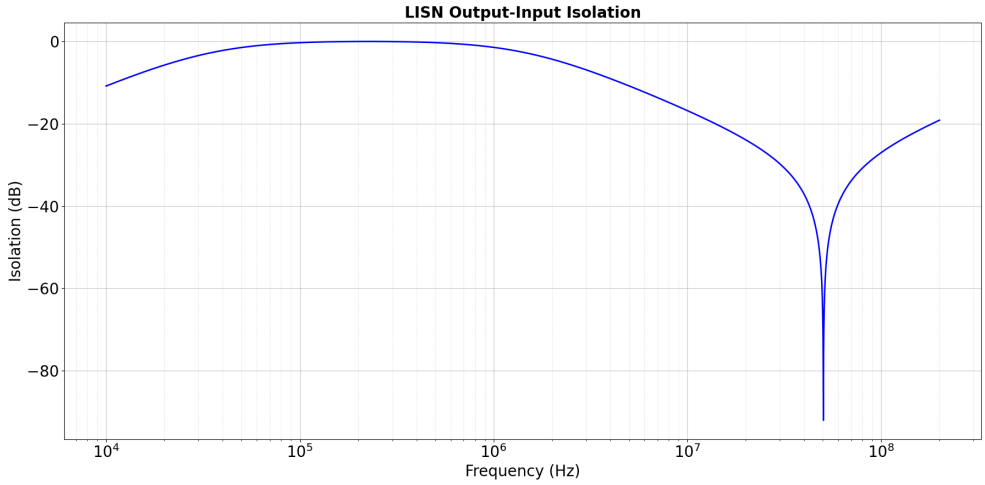


Figure 3.8: Simulated Input-to-Output Isolation response of the LISN.

The simulation results demonstrate that the LISN effectively attenuates signals propagating from the power input to the RF port throughout the operating frequency range. This characteristic minimizes the influence of supply noise on the Spectrum Analyzer input while ensuring that the signal measured at the RF port primarily consists of the conducted noise generated by the DUT.

3.4 Protection Circuit Design

After the high-frequency noise has been separated from the power line by the LISN, the signal must be conditioned before being applied to the input of the Spectrum Analyzer. During Conducted Emission measurements, the measurement instrument may be exposed to transient overvoltage pulses or high-amplitude signals generated by the DUT. In addition, the measurement signal must be AC-coupled while maintaining a stable transmission response over the entire operating frequency range.

To satisfy these requirements, an RF front-end circuit is placed between the RF port of the LISN and the Spectrum Analyzer, as shown below.

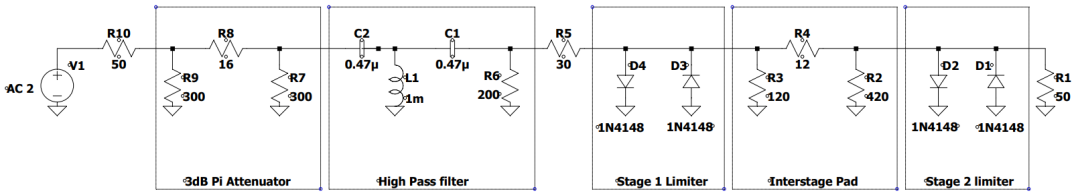


Figure 3.9: Functional blocks of the RF front-end circuit.

To evaluate the overall performance of the circuit, an AC simulation was performed using a $2\,\mathrm{V}$ excitation source, a source impedance of $50\,\Omega$, and a standard $50\,\Omega$ load representing the input impedance of the Spectrum Analyzer. The frequency sweep was carried out from $100\,\mathrm{Hz}$ to $300\,\mathrm{MHz}$.

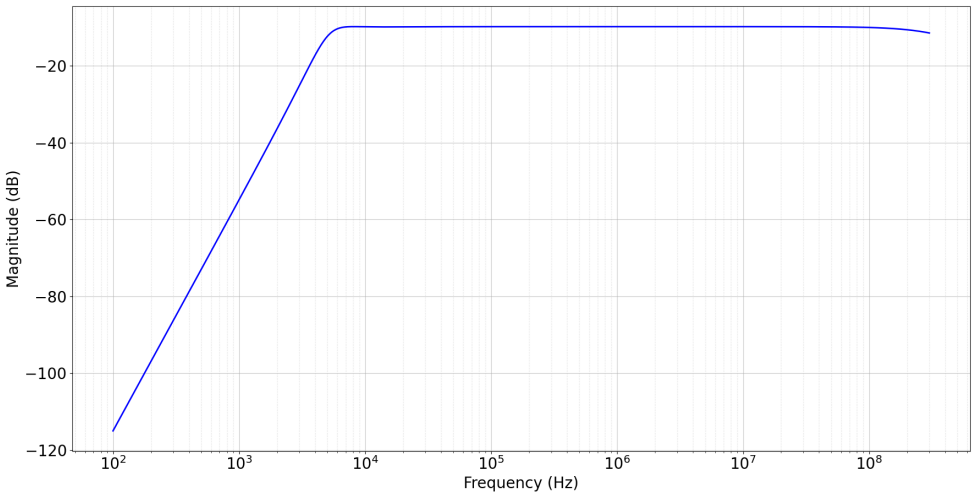


Figure 3.10: Simulated transmission response of the RF front-end circuit.

The simulation results show that the signal within the measurement frequency range is delivered to the Spectrum Analyzer with nearly constant attenuation over frequency. At low frequencies, the high-pass filter effectively suppresses the DC component and low-frequency noise, while at higher frequencies the transmission response remains stable without significant resonances within the measurement band.

Although the RF front-end introduces a certain amount of attenuation to the measurement signal, this represents a necessary trade-off to improve protection of the Spectrum Analyzer input against transient overvoltage events while also enhancing impedance matching and the stability of the frequency response. As a result, the conducted noise signal is delivered to the measurement instrument with stable characteristics, while reducing the risk of damage to the RF input during testing.

3.5 PCB Layout

After completing the schematic design and verifying the circuit through simulation, the hardware was implemented on a two-layer FR-4 printed circuit board. The components were arranged following the signal flow to minimize interconnection length and reduce the influence of parasitic elements.

The top layer is primarily used for component placement and signal routing. The power traces are designed with sufficient width to support the required current, while the RF front-end is placed close to the BNC connector to minimize transmission loss.

The bottom layer is almost entirely covered by a continuous copper ground plane. This structure provides a low-impedance return path, reduces electromagnetic interference, and improves the RF transmission characteristics of the circuit.

Figures 3.11 and 3.12 show the top and bottom PCB layouts, respectively.

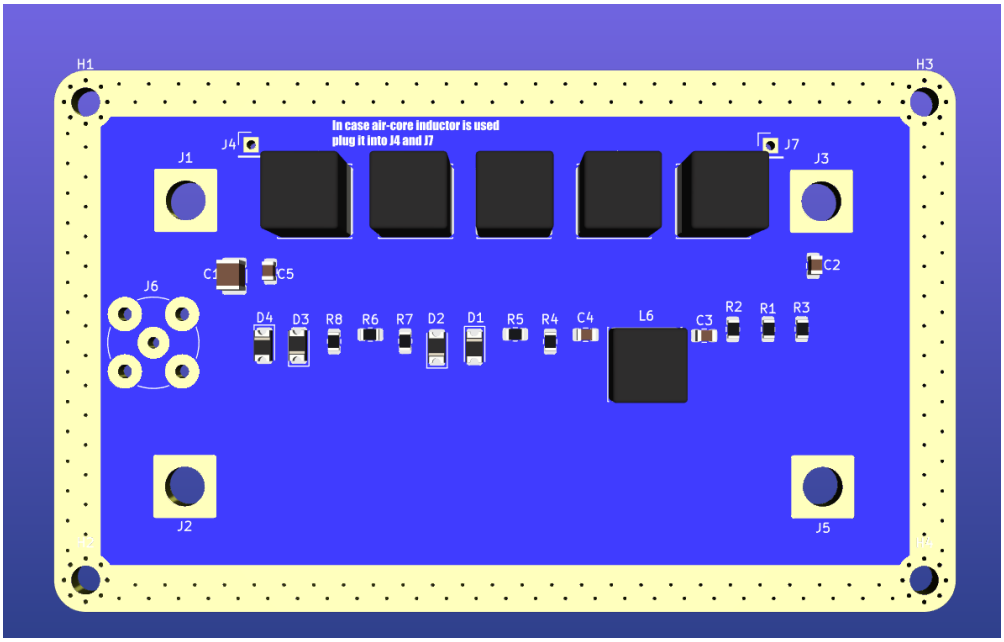


Figure 3.11: Top layer layout of the PCB.

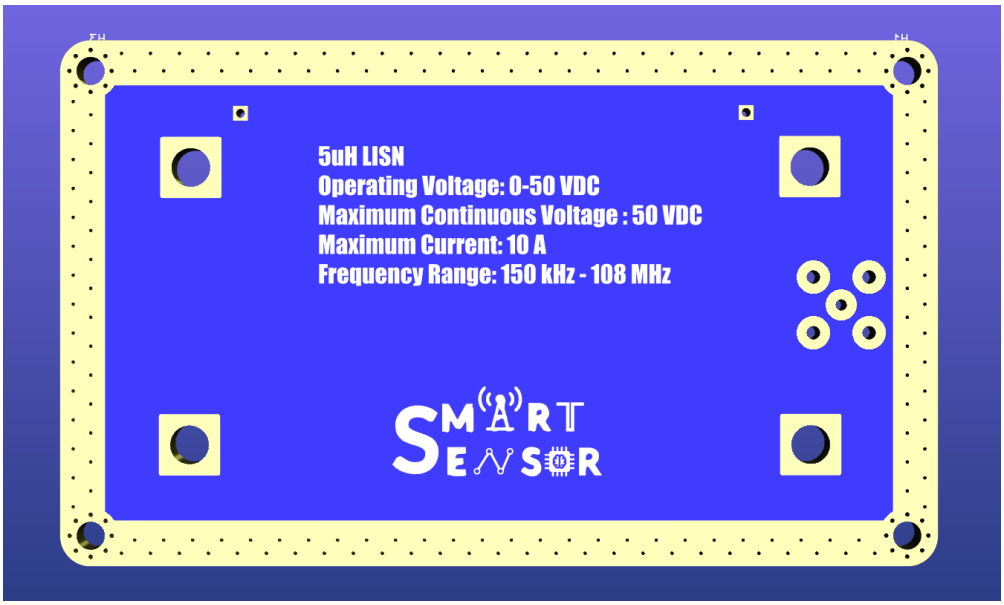


Figure 3.12: Bottom layer layout of the PCB.

3.6 Component Selection and Bill of Materials (BOM)

After completing the design and simulation stages, the next step is to select suitable components for implementing the proposed LISN. Although the schematic determines the functionality of the circuit, the actual performance of the LISN also depends on the characteristics of the individual components, including parasitic inductance, parasitic capacitance, equivalent series resistance (ESR), self-resonant frequency (SRF), and current-handling capability.

For a device operating over the frequency range from $150\,\mathrm{kHz}$ to $108\,\mathrm{MHz}$, component selection should not rely solely on nominal values. High-frequency characteristics, temperature stability, component availability, and manufacturing cost must also be taken into consideration.

In this work, the components were selected according to the following three criteria:

- Compliance with the electrical requirements specified in CISPR 25.
- Ready availability from the Vietnamese market or international distributors.
- Suitability for developing a laboratory prototype intended for research and pre-compliance testing.

The following sections describe the selection process for each component group and present the Bill of Materials (BOM) used in the proposed design.

3.6.1 Inductor

The investigation, analysis, and selection process of the inductor has already been presented in detail in Section 3.2.2.

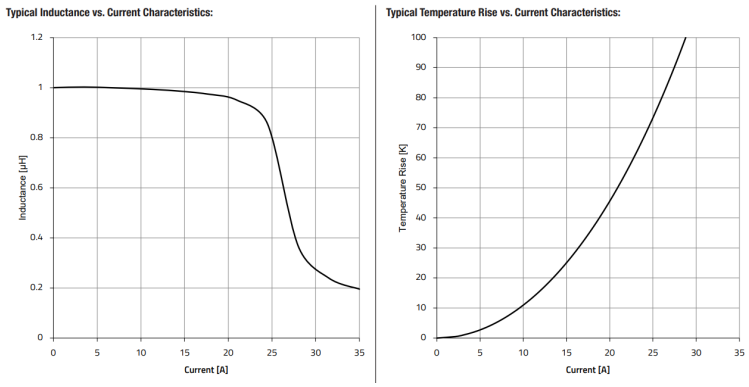


Figure 3.13: Inductance versus current and temperature rise versus current characteristics of the Würth Elektronik WE-HCC 7443340100 inductor (Source: Datasheet 7443340100 [1]).

According to the manufacturer's datasheet, the inductor exhibits a relatively stable *Inductance vs. Current* characteristic throughout its operating current range. The inductance remains nearly constant from 0 to approximately $20\,\mathrm{A}$. At the expected operating current of the proposed system (below $10\,\mathrm{A}$), the inductance is still very close to its nominal value, indicating that the magnetic core operates well below its saturation region. This behavior helps maintain the impedance characteristics of the LISN and ensures good measurement repeatability during operation [1].

Furthermore, the inductor provides a self-resonant frequency (SRF) of approximately $147\,\mathrm{MHz}$, which is higher than the upper frequency limit specified by CISPR 25 ($108\,\mathrm{MHz}$). Consequently, throughout the frequency range of interest, the inductor maintains its inductive behavior, while the influence of parasitic capacitance remains negligible [1].

In addition, the component features a maximum DC resistance (DCR) of only $3.9\,\mathrm{m\Omega}$, a rated current of $17\,\mathrm{A}$, and a saturation current of approximately $24\,\mathrm{A}$. These specifications satisfy the requirements of the LISN prototype developed in this work [1].

3.6.2 Capacitor

In this design, the capacitors used in the RF signal path were selected based on several criteria, including capacitance value, rated voltage, equivalent series resistance (ESR), equivalent series inductance (ESL), self-resonant frequency (SRF), and component availability. Compared with capacitance tolerance, parasitic parameters and high-frequency characteristics have a much greater influence on the performance of the LISN. Therefore, a tolerance of $\pm 5\%$ is considered sufficient for the proposed prototype.

Although C0G/NP0 dielectric capacitors provide superior stability and lower dielectric losses, MLCC capacitors with a C0G dielectric and a capacitance of $100\,\mathrm{nF}$ are relatively limited in terms of product availability, package options, and cost. After evaluating several alternatives, an X7R MLCC manufactured by Kyocera AVX (part number KAF21BR71H104JM) was selected for the RF coupling capacitor. This component provides a capacitance of $100\,\mathrm{nF}$, a tolerance of $\pm 5\%$, an 0805 package, and a rated voltage of $50\,\mathrm{V}$, making it suitable for the proposed prototype while remaining readily available and easy to replace during fabrication [6].

Since the conducted noise delivered to the RF measurement port carries very little power, the power-handling capability of the capacitor is not a critical design constraint. Instead, placing the capacitor close to the RF connector, minimizing PCB trace lengths, and maintaining a continuous ground plane play a much more important role in reducing parasitic inductance and preserving the high-frequency transmission characteristics of the system.

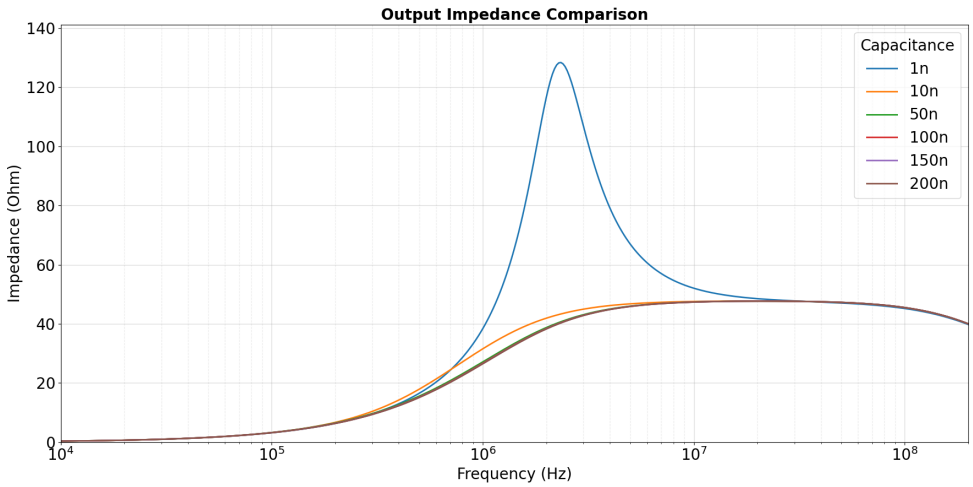


Figure 3.14: Influence of the RF coupling capacitor value on the output impedance of the LISN. Simulations were performed using capacitance values ranging from $1\,\mathrm{nF}$ to $200\,\mathrm{nF}$. The results indicate that the output impedance remains nearly unchanged throughout the frequency range of interest, demonstrating that the design is relatively insensitive to variations in the coupling capacitance.

Figure 3.14 shows that varying the RF coupling capacitor value from $1\,\mathrm{nF}$ to $200\,\mathrm{nF}$ produces only a negligible change in the output impedance of the LISN. This indicates that the circuit response is primarily determined by the $5\,\mu\mathrm{H}$ inductor and the standardized $50\,\Omega$ impedance network, whereas the coupling capacitor mainly serves to transfer the conducted noise signal to the RF measurement port while blocking the DC component.

The simulation results also demonstrate that the proposed design is highly tolerant of component variations. Therefore, the selection of a $100\,\mathrm{nF}$ capacitor is not driven by a stringent capacitance requirement but rather by considerations of component availability, voltage rating, high-frequency performance, and ease of PCB implementation.

3.7 Component Selection for the Protection and Filter Circuits

Unlike the inductor and the RF coupling capacitor, the components used in the attenuator, filter, and voltage limiter circuits do not directly contribute to establishing the standardized impedance of the LISN. Instead, their primary functions are to condition the RF signal before it reaches the spectrum analyzer and to protect the measurement instrument from abnormal voltage transients or excessive signal amplitudes.

Since these circuits do not directly determine the impedance characteristics of the LISN, component selection is primarily based on accuracy, high-frequency performance, and reliability rather than strict requirements on nominal component values.

3.7.1 Resistors

The resistors used in the attenuator and filter circuits are responsible for establishing the desired attenuation level while providing proper impedance matching between different stages of the circuit. To ensure that the attenuation closely matches the design target, thin-film resistors with a tolerance of $1\,\%$ were selected throughout both the attenuator and filter networks.

Because the RF signal at the measurement port carries very little power, the power rating of the resistors is not a critical design consideration. Standard SMD resistors rated at $0.1\,\mathrm{W}$ or $0.125\,\mathrm{W}$ are sufficient for this application. Resistors with values such as 16 Ω, 300 Ω, and others required by the design are widely available from electronic component suppliers and can be readily obtained with a tolerance of $1\,\%$.

3.7.2 Capacitors

The capacitors used in the filter circuit determine the cutoff frequency and suppress signal components outside the desired frequency band. Since these capacitors are not part of the standardized impedance network of the LISN, variations in capacitance produce only minor shifts in the filter cutoff frequency and have negligible influence on the overall performance of the system.

The proposed design employs multilayer ceramic capacitors (MLCCs) with an X7R dielectric and a tolerance of $5\,\%$. These capacitors provide low equivalent series resistance (ESR) and equivalent series inductance (ESL), occupy little PCB area, and offer satisfactory performance over the operating frequency range of the protection circuit.

In addition to their electrical characteristics, component availability was also considered during the selection process. The selected X7R MLCC capacitors are standard commercial components offered by multiple manufacturers and are readily available from electronic component distributors. This helps reduce manufacturing cost, shortens procurement time, and facilitates future reproduction and maintenance of the LISN.

3.7.3 Voltage Limiting Diodes

The limiter circuit protects the input of the spectrum analyzer from excessive voltage transients that may occur during testing or as a result of faults in the Device Under Test (DUT). Under normal operating conditions, the RF signal at the measurement port has a very small amplitude, and the diodes remain reverse-biased, introducing virtually no effect on the signal path. Only when the signal voltage exceeds the diode conduction threshold does the limiter become active, clamping the input voltage to protect the measurement instrument.

In the proposed design, the 1N4148 high-speed switching diode was selected for the limiter circuit. This device is inexpensive, widely available, and readily obtainable from commercial suppliers. Moreover, its switching time of only a few nanoseconds makes it suitable for

protection within the CISPR 25 frequency range. Although Schottky diodes or dedicated RF protection diodes offer lower junction capacitance and superior high-frequency performance, the 1N4148 adequately satisfies the requirements of the current prototype. In future revisions of the design, dedicated RF protection diodes may be considered to further minimize the influence of the limiter circuit on the measured signal.

3.8 Enclosure Design

After completing the circuit and PCB design, a dedicated enclosure was developed to protect the hardware during operation and to provide a complete mechanical structure for the LISN. The enclosure was designed according to the dimensions of the PCB and the locations of the power connectors, DUT connector, and RF connector, ensuring convenient assembly and straightforward cable connections.

In addition to protecting the circuitry from mechanical damage, the enclosure securely supports the internal components, minimizes movement during transportation, and provides sufficient space for internal wiring.

Figure 3.15 illustrates the overall appearance of the assembled LISN.

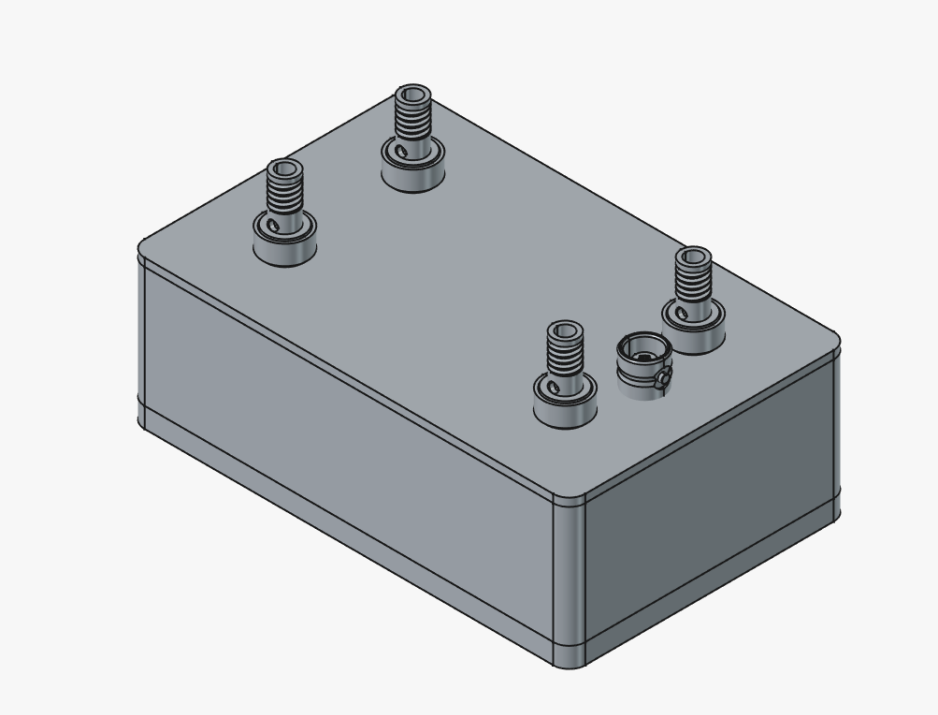


Figure 3.15: Overall view of the assembled LISN.

The internal space was arranged to accommodate the PCB, inductors, and connectors in a compact and organized manner while maintaining appropriate mechanical clearance between components. Figures 3.16 and 3.17 present the internal layout of the enclosure from two different viewing angles.

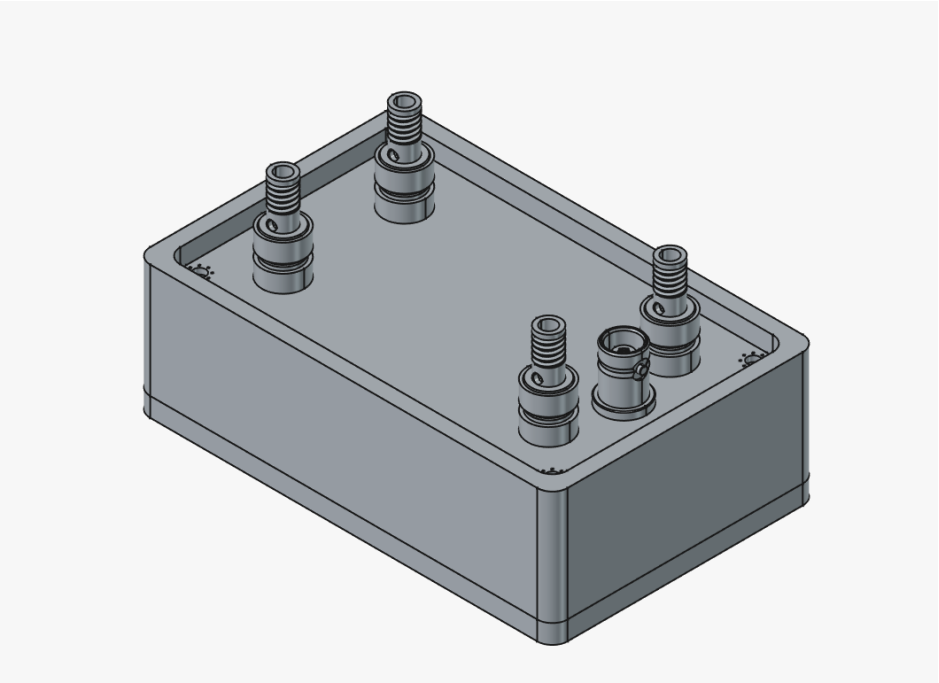


Figure 3.16: Internal component arrangement of the enclosure (first view).

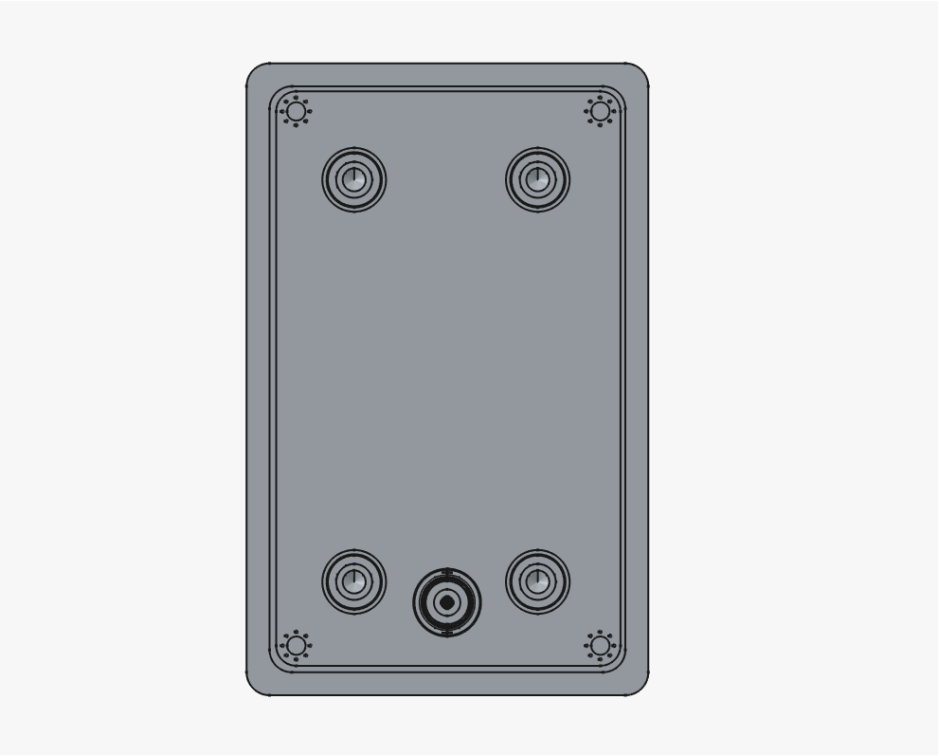


Figure 3.17: Internal component arrangement of the enclosure (second view).

The enclosure itself is illustrated in Figure 3.18. All mechanical components were designed using CAD software prior to fabrication, with particular attention paid to ease of assembly, disassembly, and maintenance.

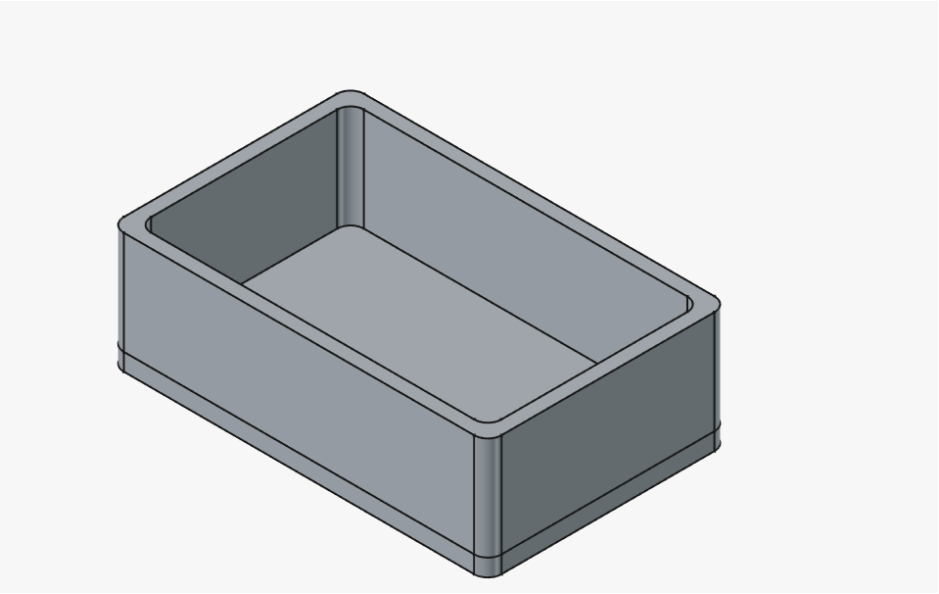


Figure 3.18: CAD model of the proposed LISN enclosure.

Chapter A Output Impedance Estimation Formula

A.1 Introduction

One of the most important characteristics of a Line Impedance Stabilization Network (LISN) is the output impedance observed from the RF measurement port. According to the CISPR 25 standard [3], the LISN must provide a nominal impedance of $50\,\Omega$ within the specified frequency range to ensure that conducted emission measurements are repeatable and comparable between different measurement systems.

During the design process, the impedance characteristic of the LISN is evaluated using AC simulation in LTspice. Instead of directly measuring the current flowing through the circuit, the impedance is derived from the voltage ratio between two measurement points in the equivalent circuit.

The expression used throughout this document is

$$Z_{out} = 50 \cdot \frac{\frac{V_{IN}}{V_{OUT}}}{1 - 2\frac{V_{IN}}{V_{OUT}}} \tag{A.1}$$

where

- V_{OUT} is the voltage at the output port of the LISN.
- V_{IN} is the voltage at the input of the Spectrum Analyzer after the standard $50\,\Omega$ termination load.

It should be noted that CISPR 25 only specifies the target impedance value of the LISN and the verification method, but does not directly define Equation (A.1). This expression is derived from the equivalent measurement model to simplify the processing of simulation results.

A.2 Equivalent Circuit

Figure A.1 shows the equivalent circuit used to determine the output impedance of the LISN.

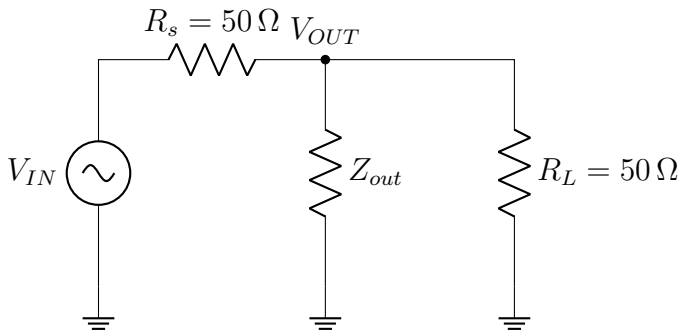


Figure A.1: Equivalent circuit used for output impedance measurement.

The excitation source is modeled as an AC source with an internal impedance of $50\,\Omega$. The input of the Spectrum Analyzer is also terminated with a standard $50\,\Omega$ load. The impedance to be determined is the output impedance of the LISN, denoted as Z_{out}.

Since the input of the measurement instrument is always terminated by a $50\,\Omega$ load, the impedance observed by the excitation source is

$$Z_{eq} = Z_{out} \parallel 50 = \frac{50 Z_{out}}{50 + Z_{out}} \tag{A.2}$$

This model is also the model used in LTspice to evaluate the impedance characteristic of the LISN.

A.3 Derivation

According to the voltage divider principle,

$$\frac{V_{IN}}{V_{OUT}} = \frac{Z_{eq}}{50 + Z_{eq}} \tag{A.3}$$

Define

$$k = \frac{V_{IN}}{V_{OUT}} \tag{A.4}$$

then

$$k = \frac{\dfrac{50 Z_{out}}{50 + Z_{out}}}{50 + \dfrac{50 Z_{out}}{50 + Z_{out}}} \tag{A.5}$$

After substituting the equivalent impedance into the voltage divider equation and simplifying, the expression for determining the LISN output impedance can be obtained as

$$Z_{out} = 50 \cdot \frac{k}{1 - 2k} \tag{A.6}$$

Substituting

$$k = \frac{V_{IN}}{V_{OUT}}$$

gives

$$Z_{out} = 50 \cdot \frac{\frac{V_{IN}}{V_{OUT}}}{1 - 2\frac{V_{IN}}{V_{OUT}}} \tag{A.7}$$

This is also the expression used in LTspice to directly calculate the impedance at each frequency point during the AC sweep.

A.4 Implementation of the Formula in LTspice

In LTspice, the above expression is directly implemented using the Trace Expression function

```
50*V(IN1)/V(OUT1)/(1-2*V(IN1)/V(OUT1))
```

The obtained result is the impedance characteristic versus frequency of the LISN, which allows evaluation of the compliance with the $50\,\Omega$ requirement according to the CISPR 25 standard.

The advantage of this method is that direct current measurement is not required during the simulation process. The entire calculation is based only on the voltage ratio between two measurement points in the equivalent circuit, simplifying data processing and allowing easy application for AC sweeps with a large number of frequency points.

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